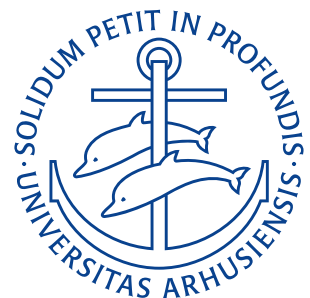


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Formula Sheet

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Course: Electromagnetism & Electronics — 290231U007
Date: January 27, 2026



Contents

1	Constant Definitions	2
1.1	Coulomb Constant	2
2	Electrostatics	2
2.1	Coulomb's Law	2
2.1.1	Principle of Superposition	2
2.2	Quantization of Charge	2
2.3	Electric Field at a point	3
2.3.1	Electric Field Produced by Point Charge	3
2.3.2	Electric Field due to a Dipole along dipole axis	3
2.3.3	Electric Field due to a Dipole along Dipole axis at Long Distance	3
2.3.4	Electric Field of a Charged Ring at Large Distance	4
2.3.5	Electric Field of a Charged Disk	4
2.3.6	Electric field from an infinite sheet	4
2.3.7	Electric field of a Conducting Sphere	5
2.4	Electric Field of an Insulating Sphere	5
2.5	Gauss' Law	6
2.5.1	Net electric flux through a surface caused by an electric field	6
2.6	Electric Potential Energy	6
2.6.1	Electrostatic Potential	7
2.6.2	Electrostatic Potential of a Plate Capacitor	7

1 Constant Definitions

1.1 Coulomb Constant

$$k = \frac{1}{4\pi\epsilon_0} = 8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2.$$

Symbols

k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

2 Electrostatics

2.1 Coulomb's Law

$$\mathbf{F} = k \frac{q_1 q_2}{r^2} \hat{r}.$$

Assumptions

- Point charges

Symbols

$\mathbf{F}$	Electrostatic force	N
k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$
q_1, q_2	Charges	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.1.1 Principle of Superposition

$$\mathbf{F}_{1,\text{net}} = \sum_{i=1}^n \mathbf{F}_{1i}.$$

Symbols

$\mathbf{F}_{1,\text{net}}$	Net force	N
$\mathbf{F}_{1i}$	Force contribution on particle 1 from particle i	N

2.2 Quantization of Charge

$$q = \pm n \cdot e.$$

Symbols

q	Charge	C
n	Integer number ($n \in \mathbb{Z}$)	—
e	Elementary charge	$1,6 \cdot 10^{-19} \text{ C}$

2.3 Electric Field at a point

$$\mathbf{E}(x, y, z) = \frac{\mathbf{F}(x, y, z)}{q_0}.$$

Symbols

$\mathbf{E}(x, y, z)$	Electric field at point (x, y, z)	$\frac{\text{N}}{\text{C}}$
$\mathbf{F}(x, y, z)$	Force on test particle at point (x, y, z)	N
q_0	Test charge	C

2.3.1 Electric Field Produced by Point Charge

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}.$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C
r	Distance	m
$\hat{r}$	Unit vector	—

2.3.2 Electric Field due to a Dipole along dipole axis

$$E(z) = E_{(+)}(z) + E_{(-)}(z) = \frac{q}{4\pi\epsilon_0 z^2} \left(\frac{1}{1 - \frac{d^2}{2z}} - \frac{1}{1 + \frac{d^2}{2z}} \right) = \frac{q}{2\pi\epsilon_0 z^3} \left(\frac{d}{\left(1 - \left(\frac{d}{2z}\right)^2\right)^2} \right).$$

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
$E_{(+)}, E_{(-)}$	Contribution to E from the positive and negative part of the dipole	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
z	Distance along axis	m
d	Separation distance	m
q	Charge	C

2.3.3 Electric Field due to a Dipole along Dipole axis at Long Distance

$$E = \frac{1}{2\pi\epsilon_0} \frac{qd}{z^3} = \frac{1}{2\pi\epsilon_0} \frac{p}{z^3}.$$

Assumptions

- $z \gg d$, i.e. the distance between charges is much smaller then the distance from the charges you are measuring ("normal" case)

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
z	Distance along dipole axis	m
d	Separation distance	m
q	Charge	C
p	Electric dipole moment ($p = qd$)	Cm

2.3.4 Electric Field of a Charged Ring at Large Distance

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2}.$$

Assumptions

- $z \gg R$, i.e. distance from the ring is much greater than the radius of the ring

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Size of charge in the ring	C
z	Distance perpendicular to the radius of the ring	m

2.3.5 Electric Field of a Charged Disk

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right).$$

Assumptions

- Infinitesimally thin disk

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
σ	Surface charge density	C/m^2
z	Distance perpendicular to the disk	m
R	radius of disk	m

2.3.6 Electric field from an infinite sheet

$$E = \frac{\sigma}{2\epsilon_0}.$$

Assumptions

- Infinitesimally thin infinite sheet

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
σ	Surface charge density	C/m^2
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

2.3.7 Electric field of a Conducting Sphere

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = k \frac{q}{r^2}.$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of a conducting sphere.
- Spherical symmetry
- No charge enclosed in conductor

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C
r	Distance	m
k	Coulomb constant	$8,99 \cdot 10^9 \text{ Nm}^2/\text{C}^2$

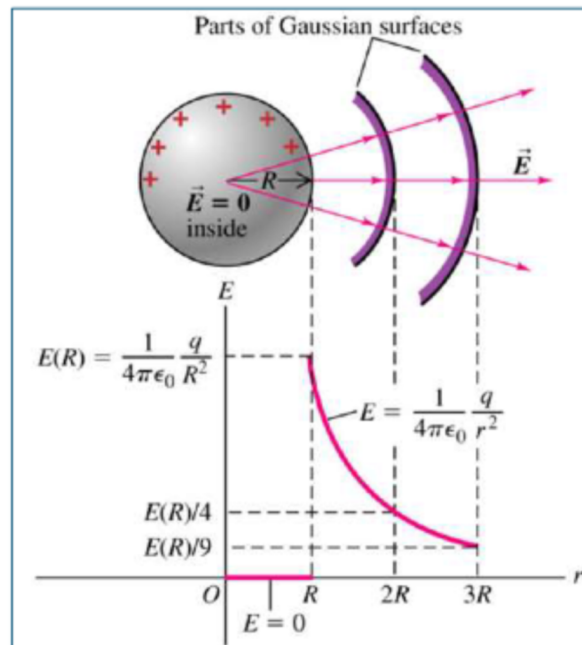


Figure 1: E-field propagation from a conducting sphere.

2.4 Electric Field of an Insulating Sphere

Outside :

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Inside :

$$E = \frac{q}{4\pi\epsilon_0 R^3} r.$$

Assumptions

- Positive electric charge Q distributed uniformly throughout the volume of an insulating sphere.
- Spherical symmetry

Symbols

E	Electric field strength	$\frac{\text{N}}{\text{C}}$
r	Distance	m
R	Radius	m
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$
q	Charge	C

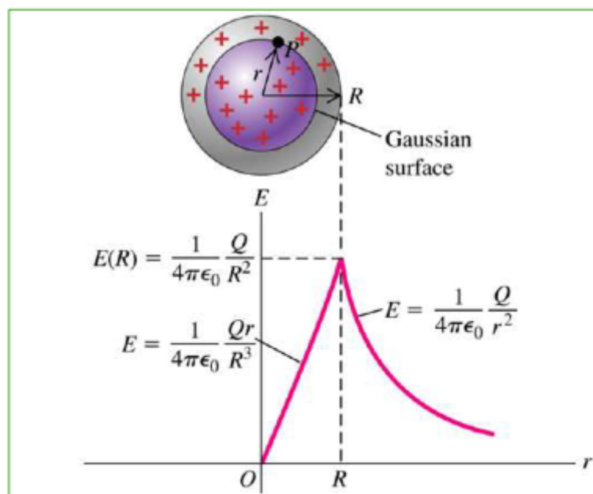


Figure 2: E-field propagation through an insulating sphere.

2.5 Gauss' Law

$$\Phi = \frac{q_{\text{enc}}}{\epsilon_0}.$$

Assumptions

- Vacuum (air)

Symbols

Φ	Electric flux	Vm
q_{enc}	Enclosed charge	C
ϵ_0	Permittivity constant	$8,85 \cdot 10^{-12} \text{ C}^2/\text{Nm}^2$

2.5.1 Net electric flux through a surface caused by an electric field

$$\Phi = \oint E \cdot dA.$$

Symbols

Φ	Electric flux	Vm
E	Electric field vector	$\frac{\text{N}}{\text{C}}$
A	Area vector	m^2

2.6 Electric Potential Energy

$$U_E = q_0 E \Delta d.$$

Symbols

U_E	Electric potential energy	J
q_0	Test charge	C
E	Electric field strength	$\frac{\text{N}}{\text{C}}$
d	Separation distance	m

2.6.1 Electrostatic Potential

$$V_E = \frac{U_E}{q_0}.$$

Symbols

V_E	Electrostatic potential	V
U_E	Electric potential energy	J
q_0	Test charge	C

2.6.2 Electrostatic Potential of a Plate Capacitor

$$V_E = Ed.$$

Symbols

V_E	Electrostatic potential	V
E	Electric field strength	$\frac{\text{N}}{\text{C}}$
d	Separation distance	m